

TRANSMISSION LINES & ELECTROMAGNETIC FIELDS

May 11, 2026

Chapter 1

Time Varying Fields and Plane Wave Propagation

1.1 Introduction

The previous discussion restricted to static, or time-invariant, EM fields. Henceforth, we shall examine situations in which electric and magnetic fields are dynamic, or time varying. It should be mentioned first that in static EM fields, electric and magnetic fields are independent of each other, whereas in dynamic EM fields, the two fields are interdependent. In other words, time-varying electric field necessarily involves a corresponding time-varying magnetic field. Second, time-varying EM fields, represented by $\mathbf{E}(x, y, z, t)$ and $\mathbf{H}(x, y, z, t)$, are of more practical value than static EM fields. However, familiarity with static fields provides a good background for understanding dynamic fields. Third, recall that electrostatic fields are usually produced by static electric charges, whereas magnetostatic fields are due to motion of electric charges with uniform velocity (direct current) or static magnetic charges (magnetic poles); time-varying fields or waves are usually due to accelerated charges or time-varying currents such as shown in Figure 1.1. Any pulsating current will produce radiation (time-varying fields). It is worth noting that pulsating current of the type shown in Figure 1.1(b) is the cause of radiated emission in digital logic boards. In summary:

stationary charges	→ electrostatic fields
steady currents	→ magnetostatic fields
time-varying currents	→ electromagnetic fields (or waves)

Our aim in this chapter is to lay a firm foundation for our subsequent studies. This will involve introducing two major concepts: (1) electromotive force based on Faraday's experiments and (2) displacement current, which resulted from Maxwell's hypothesis. As a result of these concepts, Maxwell's equations as presented in Section 7.6 and the boundary conditions for static EM fields will be modified to account for the time variation of the fields. Maxwell's equations, which summarize the laws of electromagnetism, shall be developed in this chapter.

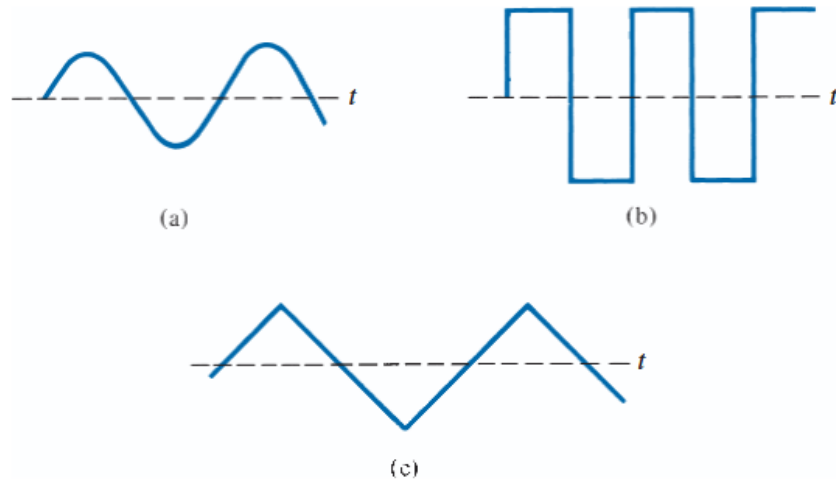


Figure 1.1: Examples of time-varying current: (a) sinusoidal, (b) rectangular, (c) triangular

Faraday's Law

After Oersted's experimental discovery (upon which Biot–Savart and Ampère based their laws) that a steady current produces a magnetic field, it seemed logical to find out whether magnetism would produce electricity. In 1831, about 11 years after Oersted's discovery, Michael Faraday in London and Joseph Henry in New York discovered that a time-varying magnetic field would produce an electric current. According to Faraday's experiments, a static magnetic field produces no current flow; but in a closed circuit, a time-varying field produces an induced voltage (called electromotive force or simply emf) that causes a flow of current.

Faraday discovered that the induced emf, V_{emf} (in volts) in any closed circuit is equal to the time rate of change of the magnetic flux linkage by the circuit.

This is called *Faraday's law*, and it can be expressed as

$$V_{\text{emf}} = -\frac{d\lambda}{dt} = -N\frac{d\Psi}{dt} \quad (1.1)$$

where $\lambda = N\Psi$ is the flux linkage, N is the number of turns in the circuit, and Ψ is the flux through each turn. The negative sign shows that the induced voltage acts in such a way as to oppose the flux producing it. This behavior is described as *Lenz's law*. Lenz's law states that the direction of current flow in the circuit is such that the induced magnetic field produced by the induced current opposes change in the original magnetic field. Recall that we described an electric field as one in which electric charges experience force. The electric fields considered so far are caused by electric charges; in such fields, the flux lines begin and end on the charges. However, electric fields of other kinds are not directly caused by electric charges. These are emf-produced fields. Sources of emf include electric generators, batteries, thermocouples, fuel cells, and photovoltaic cells, which all convert nonelectrical energy into electrical energy. Consider the electric circuit of Figure 1.2, where the battery is a source of emf. The electrochemical action of the battery results in an emf-produced field $\mathbf{E}_f$. Due to

the accumulation of charge at the battery terminals, an electrostatic field $\mathbf{E}_e (= -\nabla V)$ also exists. The total electric field at any point is

$$\mathbf{E} = \mathbf{E}_f + \mathbf{E}_e \quad (1.2)$$

Note that $\mathbf{E}_f$ is zero outside the battery, $\mathbf{E}_f$ and $\mathbf{E}_e$ have opposite directions in the battery, and the direction of $\mathbf{E}_e$ inside the battery is opposite to that outside it. If we integrate eq. (1.2) over the closed circuit, we have

$$\oint_L \mathbf{E} \cdot d\mathbf{l} = \oint_L \mathbf{E}_f \cdot d\mathbf{l} + 0 = \int_N^P \mathbf{E}_f \cdot d\mathbf{l} \quad (\text{through battery}) \quad (1.3)$$

where $\oint_L \mathbf{E}_e \cdot d\mathbf{l} = 0$ because $\mathbf{E}_e$ is conservative. The emf of the battery is the line integral of the emf-produced field, that is,

$$V_{\text{emf}} = \int_N^P \mathbf{E}_f \cdot d\mathbf{l} = - \int_N^P \mathbf{E}_e \cdot d\mathbf{l} = IR \quad (1.4)$$

since $\mathbf{E}_f$ and $\mathbf{E}_e$ are equal but opposite within the battery. It may also be regarded as the potential difference ($V_P - V_N$) between the battery's open-circuit terminals. It is important to note the following facts:

1. An electrostatic field $\mathbf{E}_e$ cannot maintain a steady current in a closed circuit, since $\oint_L \mathbf{E}_e \cdot d\mathbf{l} = 0 = IR$.
2. An emf-produced field $\mathbf{E}_f$ is nonconservative.
3. Except in electrostatics, voltage and potential difference are usually not equivalent.

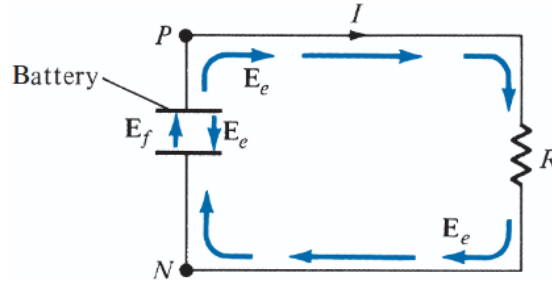


Figure 1.2: A circuit showing emf-producing field $\mathbf{E}_f$ and electrostatic field $\mathbf{E}_e$

Transformer and Motional Electromotive Forces

Having considered the connection between emf and electric field, we may examine how Faraday's law links electric and magnetic fields. For a circuit with a single turn ($N = 1$), eq. (1.1) becomes

$$V_{\text{emf}} = -\frac{d\Psi}{dt} \quad (1.5)$$

In terms of $\mathbf{E}$ and $\mathbf{B}$, eq. (1.4) can be written as

$$V_{\text{emf}} = \oint_L \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{S} \quad (1.6)$$

where Ψ has been replaced by $\int_S \mathbf{B} \cdot d\mathbf{S}$ and S is the surface area of the circuit bounded by the closed path L . It is clear from eq. (1.5) that in a time-varying situation, both electric and magnetic fields are present and are interrelated. The variation of flux with time as in eq. (1.1) or eq. (1.5) may be caused in three ways:

1. By having a stationary loop in a time-varying $\mathbf{B}$ field
2. By having a time-varying loop area in a static $\mathbf{B}$ field
3. By having a time-varying loop area in a time-varying $\mathbf{B}$ field

Each of these will be considered separately.

A. Stationary Loop in Time-Varying B Field (Transformer emf)

In Figure 1.3, a stationary conducting loop is in a time-varying magnetic $\mathbf{B}$ field. Equation (1.5) becomes

$$V_{\text{emf}} = \oint_L \mathbf{E} \cdot d\mathbf{l} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} \quad (1.7)$$

This emf induced by the time-varying current (producing the time-varying $\mathbf{B}$ field) in a stationary loop is often referred to as *transformer emf*. By applying Stokes's theorem to eq. (1.6), we obtain

$$\int_S (\nabla \times \mathbf{E}) \cdot d\mathbf{S} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} \quad (1.8)$$

For the two integrals to be equal, their integrands must be equal; that is,

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \quad (1.9)$$

This is one of Maxwell's equations for time-varying fields.

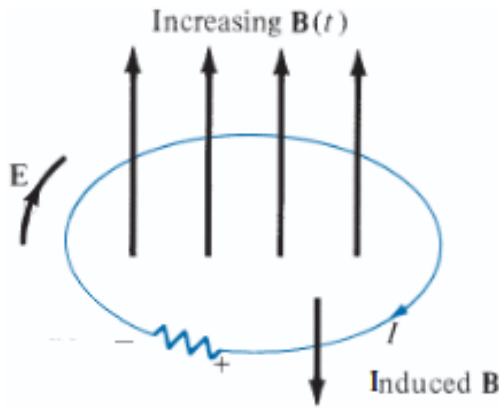


Figure 1.3: Induced emf due to a stationary loop in a time- varying B field.

B. Moving Loop in Static $\mathbf{B}$ Field (Motional emf)

When a conducting loop is moving in a static $\mathbf{B}$ field, an emf is induced in the loop. The force on a charge Q moving with uniform velocity $\mathbf{u}$ in a magnetic field $\mathbf{B}$ is

$$\mathbf{F}_m = Q \mathbf{u} \times \mathbf{B} \quad (1.10)$$

We define the motional electric field as

$$\mathbf{E}_m = \frac{\mathbf{F}_m}{Q} = \mathbf{u} \times \mathbf{B} \quad (1.11)$$

If we consider a conducting loop moving with uniform velocity $\mathbf{u}$, the emf induced in the loop is

$$V_{\text{emf}} = \oint_L \mathbf{E}_m \cdot d\mathbf{l} = \oint_L (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} \quad (1.12)$$

This type of emf is called *motional emf* or *flux-cutting emf*. Another example of motional emf is when a rod moves between a pair of rails. If $\mathbf{u}$ and $\mathbf{B}$ are perpendicular,

$$\mathbf{F}_m = I \boldsymbol{\ell} \times \mathbf{B} \quad (1.13)$$

or

$$F_m = I \ell B \quad (1.14)$$

and eq. (1.10) becomes

$$V_{\text{emf}} = u B \ell \quad (1.15)$$

By applying Stokes's theorem to eq. (1.10), we have

$$\int_S (\nabla \times \mathbf{E}_m) \cdot d\mathbf{S} = \int_S \nabla \times (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{S} \quad (1.16)$$

or

$$\nabla \times \mathbf{E}_m = \nabla \times (\mathbf{u} \times \mathbf{B}) \quad (1.17)$$

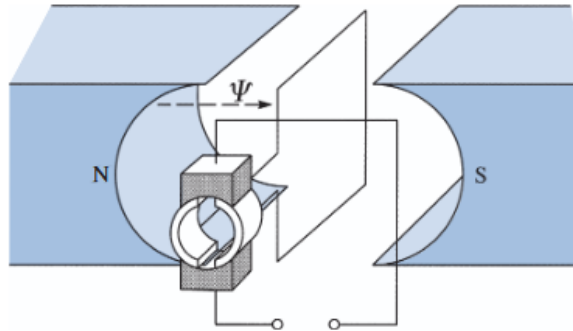


Figure 1.4: A direct-current machine

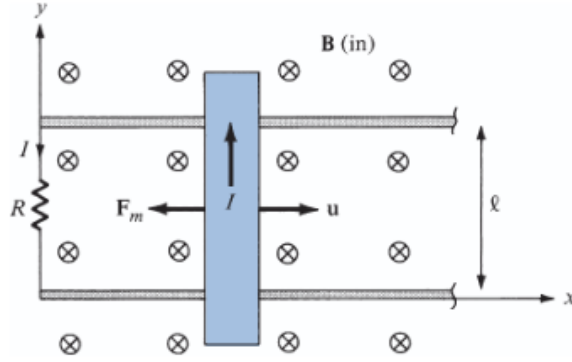


Figure 1.5: Induced emf due to a moving loop in a static B field

C. Moving Loop in Time-Varying Field

In the general case, a moving conducting loop is in a time-varying magnetic field. Both transformer emf and motional emf are present. Combining eqs. (1.6) and (1.10) gives the total emf as

$$V_{\text{emf}} = \oint_L \mathbf{E} \cdot d\mathbf{l} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{S} + \oint_L (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} \quad (1.18)$$

or from eqs. (1.8) and (1.14),

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} + \nabla \times (\mathbf{u} \times \mathbf{B}) \quad (1.19)$$

Note that eq. (1.15) is equivalent to eq. (1.4), so V_{emf} can be found using either equation.

Displacement Current

In the previous Section we have essentially reconsidered Maxwell's curl equation for electrostatic fields and modified it for time-varying situations to satisfy Faraday's law. We shall now reconsider Maxwell's curl equation for magnetic fields (Ampère's circuit law) for time-varying conditions. For static EM fields, we recall that

$$\nabla \times \mathbf{H} = \mathbf{J} \quad (1.20)$$

But the divergence of the curl of any vector field is identically zero. Hence,

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J} \quad (1.21)$$

The continuity of current, however, requires that

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \neq 0 \quad (1.22)$$

Thus eqs. (1.18) and (1.19) are incompatible for time-varying conditions. We must modify eq. (1.17) to agree with eq. (1.19). To do this, we add a term to eq. (1.17) so that it becomes

$$\nabla \times \mathbf{H} = \mathbf{J} + \mathbf{J}_d \quad (1.23)$$

where $\mathbf{J}_d$ is to be determined. Again, the divergence of the curl of any vector is zero. Hence,

$$\nabla \cdot (\nabla \times \mathbf{H}) = 0 = \nabla \cdot \mathbf{J} + \nabla \cdot \mathbf{J}_d \quad (1.24)$$

In order for eq. (1.21) to agree with eq. (1.19),

$$\nabla \cdot \mathbf{J}_d = -\nabla \cdot \mathbf{J} = \frac{\partial \rho_v}{\partial t} = \frac{\partial}{\partial t}(\nabla \cdot \mathbf{D}) = \nabla \cdot \frac{\partial \mathbf{D}}{\partial t} \quad (1.25)$$

or

$$\mathbf{J}_d = \frac{\partial \mathbf{D}}{\partial t} \quad (1.26)$$

Substituting eq. (1.22b) into eq. (1.20) results in

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \quad (1.27)$$

This is Maxwell's equation (based on Ampère's circuit law) for a time-varying field. The term $\mathbf{J}_d = \partial \mathbf{D} / \partial t$ is known as displacement current density and $\mathbf{J}$ is the conduction current density. Based on the displacement current density, we define the displacement current as

$$I_d = \int_{S_1} \mathbf{J}_d \cdot d\mathbf{S} = \int_{S_1} \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{S} \quad (1.28)$$

We must bear in mind that displacement current is a result of time-varying electric field. A typical example is the current through a capacitor when an alternating voltage source is applied to its plates. Applying an unmodified form of Ampère's circuit law to a closed path L gives

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = \int_{S_1} \mathbf{J} \cdot d\mathbf{S} = I_{\text{enc}} = I \quad (1.29)$$

where I is the current through the conductor and S_1 is the flat surface bounded by L . If we use the balloon-shaped surface S_2 that passes between the capacitor plates, we obtain

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = \int_{S_2} \mathbf{J} \cdot d\mathbf{S} = I_{\text{enc}} = 0 \quad (1.30)$$

because no conduction current ($\mathbf{J} = 0$) flows through S_2 . This is contradictory, since the same closed path L is used. To resolve the conflict, we include the displacement current in Ampère's circuit law. The total current density is $\mathbf{J} + \mathbf{J}_d$. In eq. (1.25), $\mathbf{J}_d = 0$, so the equation remains valid. In eq. (1.26), $\mathbf{J} = 0$, so that

$$\oint_L \mathbf{H} \cdot d\mathbf{l} = \int_{S_2} \mathbf{J}_d \cdot d\mathbf{S} = \frac{d}{dt} \int_{S_2} \mathbf{D} \cdot d\mathbf{S} = \frac{dQ}{dt} = I \quad (1.31)$$

So we obtain the same current for either surface, although it is conduction current in S_1 and displacement current in S_2 .

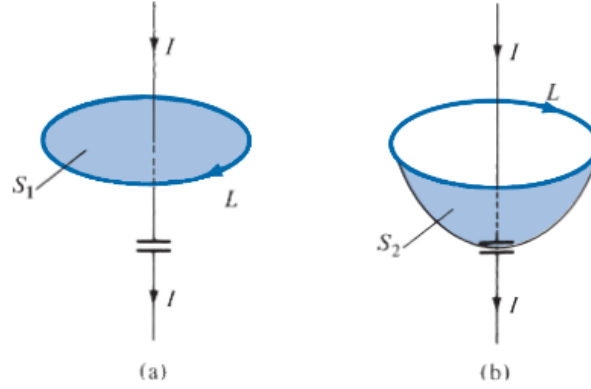


Figure 1.6: Two surfaces of integration showing the need for $\mathbf{J}d$ in Ampère's circuit law.

Maxwell's Equations in Final Forms

The Scottish physicist James Clerk Maxwell (1831–1879) is regarded as the founder of electromagnetic theory in its present form. Maxwell's celebrated work led to the discovery of electromagnetic waves. Through his theoretical efforts when he was between 35 and 40 years old, Maxwell published the first unified theory of electricity and magnetism. The theory comprised all previously known results, both experimental and theoretical, on electricity and magnetism. It further introduced displacement current and predicted the existence of electromagnetic waves. Maxwell's equations were not fully accepted by many scientists until 1888, when they were confirmed by Heinrich Rudolf Hertz (1857–1894). The German physicist was successful in generating and detecting radio waves. The laws of electromagnetism that Maxwell put together in the form of four equations were presented in Table 7.2 in Section 7.6 for static conditions. The more generalized forms of these equations are those for time-varying conditions shown in Table 9.1. We notice from the table that the divergence equations remain the same, while the curl equations have been modified. The integral form of Maxwell's equations depicts the underlying physical laws, whereas the differential form is used more frequently in solving problems. For a field to “qualify” as an electromagnetic field, it must satisfy all four Maxwell's equations. The importance of Maxwell's equations cannot be overemphasized because they summarize all known laws of electromagnetism. Since this section is meant to be a compendium of our discussion in this text, it is worthwhile to mention other equations that go hand in hand with Maxwell's equations.

Lorentz Force Equation

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{u} \times \mathbf{B}) \quad (1.32)$$

Continuity Equation

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho_v}{\partial t} \quad (1.33)$$

Constitutive Relations

$$\mathbf{D} = \varepsilon \mathbf{E} = \varepsilon_0 \mathbf{E} + \mathbf{P} \quad (1.34)$$

$$\mathbf{B} = \mu \mathbf{H} = \mu_0 (\mathbf{H} + \mathbf{M}) \quad (1.35)$$

$$\mathbf{J} = \sigma \mathbf{E} + \rho_v \mathbf{u} \quad (1.36)$$

Boundary Conditions

$$E_{1t} - E_{2t} = 0 \quad \text{or} \quad (\mathbf{E}_1 - \mathbf{E}_2) \times \mathbf{a}_n = 0 \quad (1.37)$$

$$H_{1t} - H_{2t} = K \quad \text{or} \quad (\mathbf{H}_1 - \mathbf{H}_2) \times \mathbf{a}_n = \mathbf{K} \quad (1.38)$$

$$D_{1n} - D_{2n} = \rho_s \quad \text{or} \quad (\mathbf{D}_1 - \mathbf{D}_2) \cdot \mathbf{a}_n = \rho_s \quad (1.39)$$

$$B_{1n} - B_{2n} = 0 \quad \text{or} \quad (\mathbf{B}_1 - \mathbf{B}_2) \cdot \mathbf{a}_n = 0 \quad (1.40)$$

Perfect Conductor

$$\mathbf{E} = 0, \quad \mathbf{H} = 0, \quad \mathbf{J} = 0 \quad (1.41)$$

$$B_n = 0, \quad E_t = 0 \quad (1.42)$$

Compatibility Equations

$$\nabla \cdot \mathbf{B} = -\rho_m = 0 \quad (1.43)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = \mathbf{J}_m \quad (1.44)$$

Constitutive Equations

$$\mathbf{B} = \mu \mathbf{H} \quad (1.45)$$

$$\mathbf{D} = \varepsilon \mathbf{E} \quad (1.46)$$

Equilibrium Equations

$$\nabla \cdot \mathbf{D} = \rho_v \quad (1.47)$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \quad (1.48)$$

Differential Form	Integral Form	Remarks
$\nabla \cdot \mathbf{D} = \rho_v$	$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_v \rho_v dv$	Gauss's law
$\nabla \cdot \mathbf{B} = 0$	$\oint_S \mathbf{B} \cdot d\mathbf{S} = 0$	Nonexistence of isolated magnetic charge*
$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$	$\oint_L \mathbf{E} \cdot d\mathbf{l} = -\frac{\partial}{\partial t} \int_S \mathbf{B} \cdot d\mathbf{S}$	Faraday's law
$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$	$\oint_L \mathbf{H} \cdot d\mathbf{l} = \int_S \left(\mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{S}$	Ampère's circuit law

Figure 1.7: Generalized Forms of Maxwell's Equations

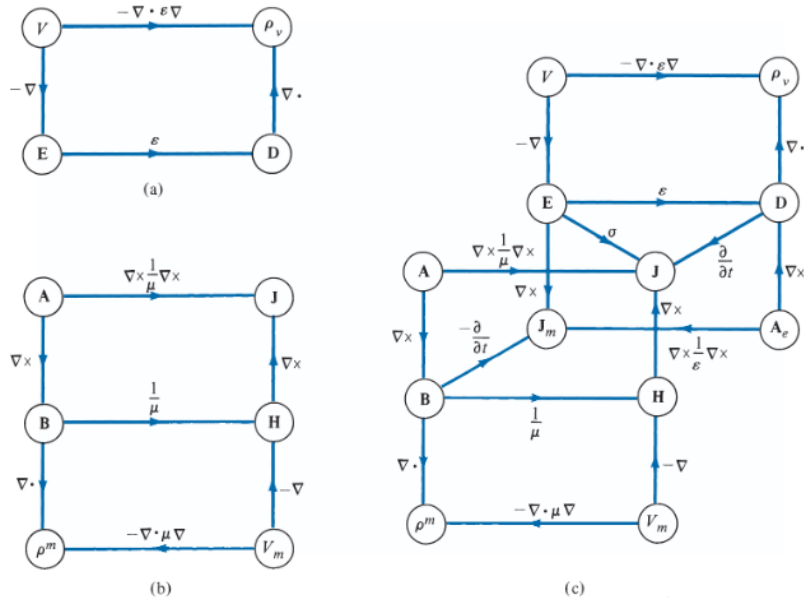


Figure 1.8: Electromagnetic flow diagrams showing the relationship between the potentials and vector fields: (a) electrostatic system, (b) magnetostatic system, (c) electromagnetic system

Time-Varying Potentials

For static EM fields, we obtained the electric scalar potential as

$$V = \int_v \frac{\rho_v dv}{4\pi\epsilon R} \quad (1.49)$$

and the magnetic vector potential as

$$\mathbf{A} = \int_v \frac{\mu \mathbf{J} dv}{4\pi R} \quad (1.50)$$

We would like to examine what happens to these potentials when the fields are time-varying. Recall that $\mathbf{A}$ was defined from the fact that $\nabla \cdot \mathbf{B} = 0$, which still holds for time-varying

fields. Hence the relation

$$\mathbf{B} = \nabla \times \mathbf{A} \quad (1.51)$$

holds for time-varying situations. Combining Faraday's law with (1.42) gives

$$\nabla \times \mathbf{E} = -\frac{\partial}{\partial t}(\nabla \times \mathbf{A}) \quad (1.52)$$

or

$$\nabla \times \left(\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} \right) = 0 \quad (1.53)$$

Thus,

$$\mathbf{E} + \frac{\partial \mathbf{A}}{\partial t} = -\nabla V \quad (1.54)$$

or

$$\mathbf{E} = -\nabla V - \frac{\partial \mathbf{A}}{\partial t} \quad (1.55)$$

Taking divergence:

$$\nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot \mathbf{A}) = -\frac{\rho_v}{\varepsilon} \quad (1.56)$$

Also,

$$\nabla^2 \mathbf{A} - \nabla(\nabla \cdot \mathbf{A}) = -\mu \mathbf{J} + \mu \varepsilon \frac{\partial}{\partial t}(\nabla V) + \mu \varepsilon \frac{\partial^2 \mathbf{A}}{\partial t^2} \quad (1.57)$$

Choosing the Lorenz condition:

$$\nabla \cdot \mathbf{A} = -\mu \varepsilon \frac{\partial V}{\partial t} \quad (1.58)$$

we obtain the wave equations:

$$\nabla^2 V - \mu \varepsilon \frac{\partial^2 V}{\partial t^2} = -\frac{\rho_v}{\varepsilon} \quad (1.59)$$

$$\nabla^2 \mathbf{A} - \mu \varepsilon \frac{\partial^2 \mathbf{A}}{\partial t^2} = -\mu \mathbf{J} \quad (1.60)$$

The solutions are:

$$V = \int_v \frac{[\rho_v]}{4\pi \varepsilon R} dv \quad (1.61)$$

$$\mathbf{A} = \int_v \frac{\mu [\mathbf{J}]}{4\pi R} dv \quad (1.62)$$

where the retarded time is

$$t' = t - \frac{R}{u} \quad (1.63)$$

and

$$u = \frac{1}{\sqrt{\mu \varepsilon}} \quad (1.64)$$

Time-Harmonic Fields

So far, our time dependence of EM fields has been arbitrary. We now assume the fields are *time harmonic*. A time-harmonic field is one that varies periodically or sinusoidally with time. A phasor is a complex number that contains the amplitude and the phase of a sinusoidal oscillation. It can be represented as

$$z = x + jy = r\angle\phi \quad (1.65)$$

or

$$z = re^{j\phi} = r(\cos\phi + j\sin\phi) \quad (1.66)$$

where

$$r = |z| = \sqrt{x^2 + y^2} \quad (1.67)$$

and

$$\phi = \tan^{-1}\left(\frac{y}{x}\right) \quad (1.68)$$

Basic operations: **Addition**

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2) \quad (1.69)$$

Subtraction

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2) \quad (1.70)$$

Multiplication

$$z_1 z_2 = r_1 r_2 \angle(\phi_1 + \phi_2) \quad (1.71)$$

Division

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \angle(\phi_1 - \phi_2) \quad (1.72)$$

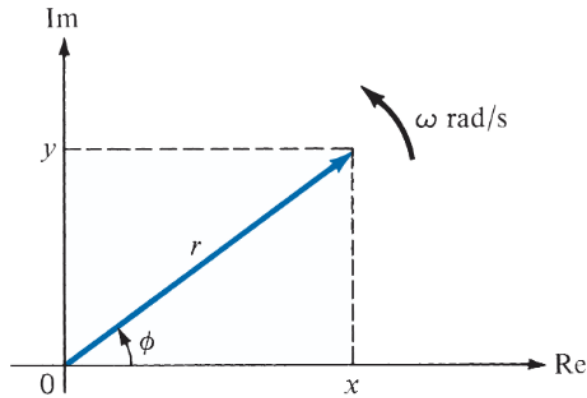


Figure 1.9: Representation of a phasor $z = x + jy = r\angle\phi$.

Square root:

$$\sqrt{z} = \sqrt{r} \angle \frac{\phi}{2} \quad (1.73)$$

Complex conjugate:

$$z^* = x - jy = r\angle(-\phi) = re^{-j\phi} \quad (1.74)$$

Other properties of complex numbers can be found in Appendix A.2. To introduce the time element, we let

$$\phi = \omega t + \theta \quad (1.75)$$

where θ may be a function of time or space coordinates or a constant. The real (Re) and imaginary (Im) parts of

$$re^{j\phi} = re^{j(\omega t + \theta)} = re^{j\theta}e^{j\omega t} \quad (1.76)$$

are respectively given by

$$\text{Re}\{re^{j\phi}\} = r \cos(\omega t + \theta) \quad (1.77)$$

$$\text{Im}\{re^{j\phi}\} = r \sin(\omega t + \theta) \quad (1.78)$$

Thus, a sinusoidal current

$$I(t) = I_0 \cos(\omega t + \theta)$$

equals the real part of $I_0 e^{j\theta} e^{j\omega t}$. The complex term $I_0 e^{j\theta}$ is called the phasor current:

$$I_s = I_0 e^{j\theta} = I_0 \angle \theta \quad (1.79)$$

Thus, the instantaneous current can be written as

$$I(t) = \text{Re}\{I_s e^{j\omega t}\} \quad (1.80)$$

In general, a phasor may represent a scalar or vector quantity. If $\mathbf{A}(x, y, z, t)$ is time-harmonic, then

$$\mathbf{A}(x, y, z, t) = \text{Re}\{\mathbf{A}_s(x, y, z) e^{j\omega t}\} \quad (1.81)$$

Example:

$$\mathbf{A} = \text{Re}\{A_0 e^{-j\beta x} \mathbf{a}_y e^{j\omega t}\} \quad (1.82)$$

Thus the phasor form is

$$\mathbf{A}_s = A_0 e^{-j\beta x} \mathbf{a}_y \quad (1.83)$$

From (1.67),

$$\frac{\partial \mathbf{A}}{\partial t} = \text{Re}\{j\omega \mathbf{A}_s e^{j\omega t}\} \quad (1.84)$$

Thus,

$$\frac{\partial \mathbf{A}}{\partial t} \rightarrow j\omega \mathbf{A}_s \quad (1.85)$$

Similarly,

$$\int \mathbf{A} dt \rightarrow \frac{\mathbf{A}_s}{j\omega} \quad (1.86)$$

Example using Maxwell's equation:

$$\nabla \times \mathbf{E}(x, y, z, t) = -\frac{\partial}{\partial t} \mathbf{B}(x, y, z, t) \quad (1.87)$$

Let

$$\mathbf{E}(x, y, z, t) = \text{Re}\{\mathbf{E}_s(x, y, z) e^{j\omega t}\}$$

$$\mathbf{B}(x, y, z, t) = \text{Re}\{\mathbf{B}_s(x, y, z)e^{j\omega t}\}$$

Substituting,

$$\nabla \times \text{Re}\{\mathbf{E}_s e^{j\omega t}\} = -\frac{\partial}{\partial t} \text{Re}\{\mathbf{B}_s e^{j\omega t}\} \quad (1.88)$$

Left-hand side:

$$\nabla \times \{\text{Re}[\mathbf{E}_s e^{j\omega t}]\} = \text{Re}\{(\nabla \times \mathbf{E}_s) e^{j\omega t}\} \quad (1.89)$$

Right-hand side:

$$-\frac{\partial}{\partial t} \{\text{Re}[\mathbf{B}_s e^{j\omega t}]\} = -\text{Re}\{j\omega \mathbf{B}_s e^{j\omega t}\} \quad (1.90)$$

Comparing:

$$\nabla \times \mathbf{E}_s = -j\omega \mathbf{B}_s \quad (1.91)$$

Point Form	Integral Form
$\nabla \cdot \mathbf{D}_s = \rho_{vs}$	$\oint \mathbf{D}_s \cdot d\mathbf{S} = \int \rho_{vs} dv$
$\nabla \cdot \mathbf{B}_s = 0$	$\oint \mathbf{B}_s \cdot d\mathbf{S} = 0$
$\nabla \times \mathbf{E}_s = -j\omega \mathbf{B}_s$	$\oint \mathbf{E}_s \cdot d\mathbf{l} = -j\omega \int \mathbf{B}_s \cdot d\mathbf{S}$
$\nabla \times \mathbf{H}_s = \mathbf{J}_s + j\omega \mathbf{D}_s$	$\oint \mathbf{H}_s \cdot d\mathbf{l} = \int (\mathbf{J}_s + j\omega \mathbf{D}_s) \cdot d\mathbf{S}$

Figure 1.10: Time-Harmonic Maxwell's Equations Assuming Time Factor $e(j\omega t)$

ELECTROMAGNETIC WAVE PROPAGATION

Our first application of Maxwell's equations will be in relation to electromagnetic wave propagation. The existence of EM waves, predicted by Maxwell's equations, was first investigated by Heinrich Hertz. After several calculations and experiments, Hertz succeeded in generating and detecting radio waves, which are sometimes called Hertzian waves in his honor.

In general, waves are means of transporting energy or information.

Typical examples of EM waves include radio waves, TV signals, radar beams, and light rays. All forms of EM energy share three fundamental characteristics: they all travel at high velocity; in traveling, they assume the properties of waves; and they radiate outward from a source, without benefit of any discernible physical vehicles. The problem of radiation will be addressed in Chapter 13. In this chapter, our major goal is to solve Maxwell's equations and describe EM wave motion in the following media:

1. Free space ($\sigma = 0$, $\varepsilon = \varepsilon_0$, $\mu = \mu_0$)
2. Lossless dielectrics ($\sigma \simeq 0$, $\varepsilon = \varepsilon_r \varepsilon_0$, $\mu = \mu_r \mu_0$, or $\sigma \ll \omega \varepsilon$)
3. Lossy dielectrics ($\sigma \neq 0$, $\varepsilon = \varepsilon_r \varepsilon_0$, $\mu = \mu_r \mu_0$)

4. Good conductors ($\sigma \simeq \infty$, $\varepsilon = \varepsilon_0$, $\mu = \mu_r \mu_0$, or $\sigma \gg \omega \varepsilon$)

where ω is the angular frequency of the wave. Case 3, for lossy dielectrics, is the most general case and will be considered first. Once this general case has been solved, we simply derive the other cases (1, 2, and 4) from it as special cases by changing the values of σ , μ , and ε . However, before we consider wave motion in those different media, it is appropriate that we study the characteristics of waves in general. This is important for proper understanding of EM waves. The reader who is conversant with the concept of waves may skip Section 10.2. Power considerations, reflection, and transmission between two different media will be discussed later in the chapter.

WAVES IN GENERAL

A clear understanding of EM wave propagation depends on a grasp of what waves are in general.

A wave is a function of both space and time.

Wave motion occurs when a disturbance at point A , at time t_0 , is related to what happens at point B , at time $t > t_0$. A wave equation, as exemplified by eqs. (1.51) and (1.52), is a partial differential equation of the second order. In one dimension, a scalar wave equation takes the form

$$\frac{\partial^2 E}{\partial t^2} - u^2 \frac{\partial^2 E}{\partial z^2} = 0 \quad (1.92)$$

where u is the *wave velocity*. Equation (10.1) is a special case of eq. (1.51) in which the medium is source free ($\rho_v = J = 0$). It can be solved by following a procedure similar to that in Example 6.5. Its solutions are of the form

$$E^+ = f(z - ut) \quad (1.93)$$

or

$$E^- = g(z + ut) \quad (1.94)$$

$$E = f(z - ut) + g(z + ut) \quad (1.95)$$

where f and g denote any function of $z - ut$ and $z + ut$, respectively. Examples of such functions include $z \pm ut$, $\sin k(z \pm ut)$, $\cos k(z \pm ut)$, and $e^{jk(z \pm ut)}$, where k is a constant. It can easily be shown that these functions all satisfy eq. (10.1). If we particularly assume harmonic (or sinusoidal) time dependence $e^{j\omega t}$, eq. (10.1) becomes

$$\frac{d^2 E_s}{dz^2} + \beta^2 E_s = 0 \quad (1.96)$$

where $\beta = \omega/u$ and E_s is the phasor form of E . The solution to eq. (10.3) is similar to Case 3 of Example 6.5. With the time factor inserted, the possible solutions to eq. (10.3) are

$$E^+ = A e^{j(\omega t - \beta z)} \quad (1.97)$$

$$E^- = B e^{j(\omega t + \beta z)} \quad (1.98)$$

where E^+ means positive z -travel and E^- means negative travel. Combining E^+ and E^- leads to

$$E = Ae^{j(\omega t - \beta z)} + Be^{j(\omega t + \beta z)} \quad (1.99)$$

where A and B are real constants. For the moment, let us consider the solution in eq. (10.4a). Taking the imaginary part of this equation, we have

$$E = A \sin(\omega t - \beta z) \quad (1.100)$$

This is a sine wave chosen for simplicity; a cosine wave would have resulted had we taken the real part of eq. (10.4a). Note the following characteristics of the wave in eq. (10.5):

1. It is time harmonic because we assumed time dependence of the form $e^{j\omega t}$ to arrive at eq. (10.5).
2. The amplitude of the wave A has the same units as E .
3. The phase (in radians) of the wave depends on time t and space variable z , it is the term $(\omega t - \beta z)$.
4. The angular frequency ω is given in radians per second; β , the *phase constant* or *wave number*, is given in radians per meter.

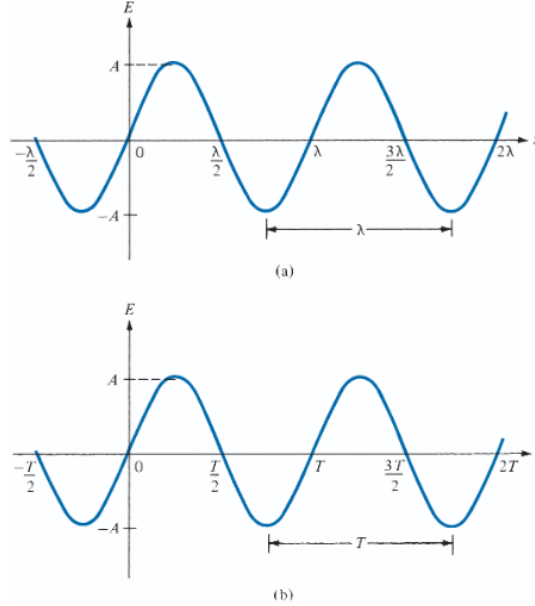


Figure 1.11: Plot of $E(z, t) = A \sin(\omega t - \beta z)$: (a) with constant t , (b) with constant z .

Because E varies with both time t and the space variable z , we may plot E as a function of t by keeping z constant and vice versa. The plots of $E(z, t = \text{constant})$ and $E(t, z = \text{constant})$ are shown in Figure 10.1(a) and (b), respectively. From Figure 10.1(a), we observe that the wave takes distance λ to repeat itself and hence λ is called the wavelength (in meters). From Figure 10.1(b), the wave takes time T to repeat itself; consequently T is known as the period in seconds. Since it takes time T for the wave to travel distance λ at the speed u , we expect

$$\lambda = uT \quad (1.101)$$

But $T = 1/f$, where f is the frequency (the number of cycles per second) of the wave in hertz (Hz). Hence,

$$u = f\lambda \quad (1.102)$$

Because of this fixed relationship between wavelength and frequency, one can identify the position of a radio station with its band by either the frequency or the wavelength. Usually the frequency is preferred. Also, because

$$\omega = 2\pi f \quad (1.103)$$

$$\beta = \frac{\omega}{u} \quad (1.104)$$

and

$$T = \frac{1}{f} = \frac{2\pi}{\omega} \quad (1.105)$$

we expect from eqs. (10.6) and (10.7) that

$$\beta = \frac{2\pi}{\lambda} = \frac{\omega}{u} \quad (1.106)$$

Equation (10.8) shows that for every wavelength of distance traveled, a wave undergoes a phase change of 2π radians. We will now show that the wave represented by eq. (10.5) is traveling with a velocity u in the $+z$ -direction. To do this, we consider a fixed point P on the wave. Point P is a point of constant phase, therefore

$$\omega t - \beta z = \text{constant}$$

or

$$\frac{dz}{dt} = \frac{\omega}{\beta} = u \quad (1.107)$$

which is the same as eq. (10.7b). Equation (10.9) shows that the wave travels with velocity u in the $+z$ -direction. Similarly, it can be shown that the wave $B \sin(\omega t + \beta z)$ in eq. (10.4b) is traveling with velocity u in the $-z$ -direction. In summary, we note the following:

1. A wave is a function of both time and space.
2. Though time $t = 0$ is arbitrarily selected as a reference for the wave, a wave is without beginning or end.
3. A negative sign in $(\omega t \pm \beta z)$ is associated with a wave propagating in the $+z$ -direction (forward-traveling or positive-going wave), whereas a positive sign indicates that a wave is traveling in the $-z$ -direction (backward-traveling or negative-going wave).
4. Since $\sin(-\psi) = -\sin \psi$, whereas $\cos(-\psi) = \cos \psi$,

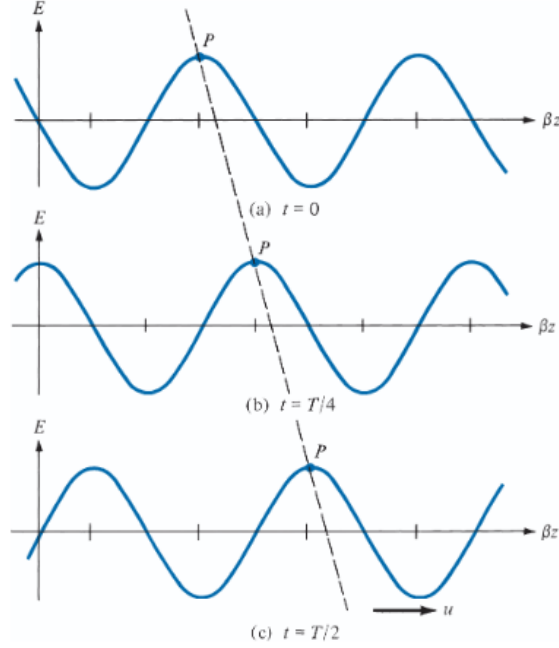


Figure 1.12: Plot of $E(z, t) = A \sin(\omega t - \beta z)$ at time (a) $t = 0$, (b) $t = T/4$, (c) $t = T/2$; P moves in the $+z$ -direction with velocity u .

$$\sin(\psi \pm \pi/2) = \pm \cos \psi \quad (1.108)$$

$$\sin(\psi \pm \pi) = -\sin \psi \quad (1.109)$$

$$\cos(\psi \pm \pi/2) = \mp \sin \psi \quad (1.110)$$

$$\cos(\psi \pm \pi) = -\cos \psi \quad (1.111)$$

where $\psi = \omega t \pm \beta z$. One of the relations in eqs. (10.10) can be used to represent any time-harmonic wave in the form of sine or cosine.

5. E and H are called uniform waves if they lie in a plane and are constant over such planes.

A large number of frequencies visualized in numerical order constitute a *spectrum*. Table 10.1 shows the frequencies at which various types of energy in the EM spectrum occur. Frequencies used for radio communication occur near the lower end of the EM spectrum. As frequency increases, the manifestation of EM energy becomes dangerous to human beings. Microwave ovens, for example, can pose a hazard if not properly shielded. As the practical difficulties of using EM energy for communication purposes also increase as frequency increases, until finally it can no longer be used. As communication methods improve, the limit to usable frequency has been pushed higher. Today communication satellites use frequencies near 14 GHz. This is still far below light frequencies, but in the enclosed environment of fiber optics, light itself can be used for radio communication.

EM Phenomena	Examples of Uses	Approximate Frequency Range
Cosmic rays	Physics, astronomy	10^{14} GHz and above
Gamma rays	Cancer therapy	10^{10} – 10^{13} GHz
X-rays	X-ray examination	10^8 – 10^9 GHz
Ultraviolet radiation	Sterilization	10^6 – 10^8 GHz
Visible light	Human vision	10^5 – 10^6 GHz
Infrared radiation	Photography	10^3 – 10^4 GHz
Microwave waves	Radar, microwave relays, satellite communication	3–300 GHz
Radio waves	UHF television	470–806 MHz
	VHF television, FM radio	54–216 MHz
	Short-wave radio	3–26 MHz
	AM radio	535–1605 kHz

Figure 1.13: Electromagnetic Spectrum

WAVE PROPAGATION IN LOSSY DIELECTRICS

As mentioned in Section 10.1, wave propagation in lossy dielectrics is a general case from which wave propagation in media of other types can be derived as special cases. Therefore, this section is foundational to the next three sections.

A lossy dielectric is a medium in which an EM wave, as it propagates, loses power owing to imperfect dielectric.

In other words, a lossy dielectric is a partially conducting medium (imperfect dielectric or imperfect conductor) with $\sigma \neq 0$, as distinct from a lossless dielectric (perfect or good dielectric) in which $\sigma = 0$. Consider a linear, isotropic, homogeneous, lossy dielectric medium that is charge free (macroscopic $\rho_v = 0$). Assuming and suppressing the time factor $e^{j\omega t}$, Maxwell's equations become

$$\nabla \cdot \mathbf{E}_s = 0 \quad (1.112)$$

$$\nabla \cdot \mathbf{H}_s = 0 \quad (1.113)$$

$$\nabla \times \mathbf{E}_s = -j\omega\mu\mathbf{H}_s \quad (1.114)$$

$$\nabla \times \mathbf{H}_s = (\sigma + j\omega\varepsilon)\mathbf{E}_s \quad (1.115)$$

Taking the curl of both sides of eq. (10.13) gives

$$\nabla \times \nabla \times \mathbf{E}_s = -j\omega\mu(\nabla \times \mathbf{H}_s) \quad (1.116)$$

Applying the vector identity

$$\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \quad (1.117)$$

and invoking eqs. (10.11) and (10.14), we obtain

$$(\nabla \nabla \cdot \mathbf{E}_s) - \nabla^2 \mathbf{E}_s = -j\omega\mu(\sigma + j\omega\varepsilon)\mathbf{E}_s$$

or

$$\nabla^2 \mathbf{E}_s - \gamma^2 \mathbf{E}_s = 0 \quad (1.118)$$

where

$$\gamma^2 = j\omega\mu(\sigma + j\omega\varepsilon) \quad (1.119)$$

Similarly, for $\mathbf{H}$ field,

$$\nabla^2 \mathbf{H}_s - \gamma^2 \mathbf{H}_s = 0 \quad (1.120)$$

Since γ is complex, we may let

$$\gamma = \alpha + j\beta \quad (1.121)$$

From eqs. (10.18) and (10.20):

$$-\text{Re } \gamma^2 = \beta^2 - \alpha^2 = \omega^2 \mu \varepsilon \quad (1.122)$$

$$|\gamma^2| = \beta^2 + \alpha^2 = \omega \mu \sqrt{\sigma^2 + \omega^2 \varepsilon^2} \quad (1.123)$$

Hence,

$$\alpha = \omega \sqrt{\frac{\mu \varepsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \varepsilon} \right)^2} - 1 \right]} \quad (1.124)$$

$$\beta = \omega \sqrt{\frac{\mu \varepsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \varepsilon} \right)^2} + 1 \right]} \quad (1.125)$$

Without loss of generality, assume propagation along $+a_z$ and

$$\mathbf{E}_s = E_{xs}(z) \mathbf{a}_x \quad (1.126)$$

Substituting into eq. (10.17):

$$(\nabla^2 - \gamma^2) E_{xs}(z) = 0 \quad (1.127)$$

which reduces to

$$\left(\frac{d^2}{dz^2} - \gamma^2 \right) E_{xs}(z) = 0 \quad (1.128)$$

The solution is

$$E_{xs}(z) = E_0 e^{-\gamma z} + E'_0 e^{\gamma z} \quad (1.129)$$

For wave propagation along $+a_z$, $E'_0 = 0$. Inserting the time factor,

$$\mathbf{E}(z, t) = \text{Re} [E_0 e^{-\gamma z} e^{j\omega t} \mathbf{a}_x]$$

or

$$\mathbf{E}(z, t) = E_0 e^{-\alpha z} \cos(\omega t - \beta z) \mathbf{a}_x \quad (1.130)$$

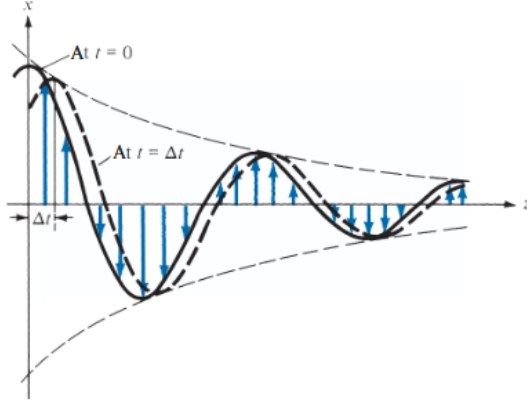


Figure 1.14: Enter Caption

Similarly,

$$\mathbf{H}(z, t) = \text{Re} [H_0 e^{-\gamma z} e^{j\omega t} \mathbf{a}_y] \quad (1.131)$$

where

$$H_0 = \frac{E_0}{\eta} \quad (1.132)$$

and the intrinsic impedance is

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\varepsilon}} = |\eta| e^{j\theta_\eta} \quad (1.133)$$

with

$$|\eta| = \frac{\sqrt{\mu/\varepsilon}}{\left[1 + \left(\frac{\sigma}{\omega\varepsilon}\right)^2\right]^{1/4}}, \quad \tan 2\theta_\eta = \frac{\sigma}{\omega\varepsilon} \quad (1.134)$$

Hence,

$$\mathbf{H} = \frac{E_0}{|\eta|} e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta) \mathbf{a}_y \quad (1.135)$$

The attenuation in nepers per meter:

$$1 \text{ Np} = 20 \log_{10} e = 8.686 \text{ dB} \quad (1.136)$$

Wave velocity and wavelength:

$$u = \frac{\omega}{\beta}, \quad \lambda = \frac{2\pi}{\beta} \quad (1.137)$$

The ratio of conduction to displacement current:

$$\tan \theta = \frac{\sigma}{\omega\varepsilon} \quad (1.138)$$

and

$$\theta = 2\theta_\eta \quad (1.139)$$

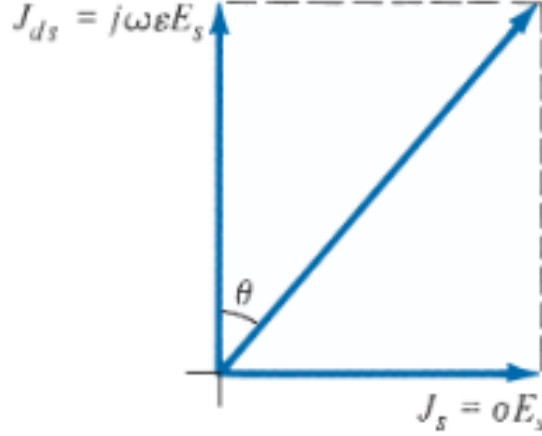


Figure 1.15: Loss angle of a lossy medium.

From eq. (10.14),

$$\nabla \times \mathbf{H}_s = (\sigma + j\omega\epsilon)\mathbf{E}_s = j\omega\epsilon_c\mathbf{E}_s \quad (1.140)$$

where

$$\epsilon_c = \epsilon \left(1 - j\frac{\sigma}{\omega\epsilon}\right) \quad (1.141)$$

or

$$\epsilon_c = \epsilon' - j\epsilon'' \quad (1.142)$$

Hence,

$$\tan \theta = \frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega\epsilon} \quad (1.143)$$

PLANE WAVES IN LOSSLESS DIELECTRICS

In a lossless dielectric, $\sigma \ll \omega\epsilon$. It is a special case of that in Section 10.3 except that

$$\sigma = 0, \quad \epsilon = \epsilon_0\epsilon_r, \quad \mu = \mu_0\mu_r \quad (1.144)$$

Substituting these into eqs. (10.23) and (10.24) gives

$$\alpha = 0, \quad \beta = \omega\sqrt{\mu\epsilon} \quad (1.145)$$

$$u = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu\epsilon}}, \quad \lambda = \frac{2\pi}{\beta} \quad (1.146)$$

Also,

$$\eta = \sqrt{\frac{\mu}{\epsilon}} \angle 0^\circ \quad (1.147)$$

and thus $\mathbf{E}$ and $\mathbf{H}$ are in time phase with each other.

PLANE WAVES IN FREE SPACE

Plane waves in free space comprise a special case of what we considered in Section 10.3. In this case,

$$\sigma = 0, \quad \varepsilon = \varepsilon_0, \quad \mu = \mu_0 \quad (1.148)$$

This may also be regarded as a special case of Section 10.4. Thus we simply replace ε by ε_0 and μ by μ_0 in eq. (10.43), or we substitute eq. (10.45) directly into eqs. (10.23) and (10.24). Either way, we obtain

$$\alpha = 0, \quad \beta = \omega\sqrt{\mu_0\varepsilon_0} = \frac{\omega}{c} \quad (1.149)$$

$$u = \frac{1}{\sqrt{\mu_0\varepsilon_0}} = c, \quad \lambda = \frac{2\pi}{\beta} \quad (1.150)$$

where $c \simeq 3 \times 10^8$ m/s, the speed of light in a vacuum. The fact that EM waves travel in free space at the speed of light is significant. It provides some evidence that light is the manifestation of an EM wave. In other words, light is characteristically electromagnetic. By substituting the constitutive parameters in eq. (10.45) into eq. (10.33), $\theta_\eta = 0$ and $\eta = \eta_0$, where η_0 is called the intrinsic impedance of free space and is given by

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 120\pi \approx 377 \Omega \quad (1.151)$$

$$\mathbf{E} = E_0 \cos(\omega t - \beta z) \mathbf{a}_x \quad (1.152)$$

then

$$\mathbf{H} = H_0 \cos(\omega t - \beta z) \mathbf{a}_y = \frac{E_0}{\eta_0} \cos(\omega t - \beta z) \mathbf{a}_y \quad (1.153)$$

The plots of $\mathbf{E}$ and $\mathbf{H}$ are shown in Figure 10.7(a). In general, if $\mathbf{a}_E$, $\mathbf{a}_H$, and $\mathbf{a}_k$ are unit vectors along the E field, the H field, and the direction of wave propagation, it can be shown that

$$\mathbf{a}_k \times \mathbf{a}_E = \mathbf{a}_H$$

or

$$\mathbf{a}_k \times \mathbf{a}_H = -\mathbf{a}_E$$

or

$$\mathbf{a}_E \times \mathbf{a}_H = \mathbf{a}_k \quad (1.154)$$

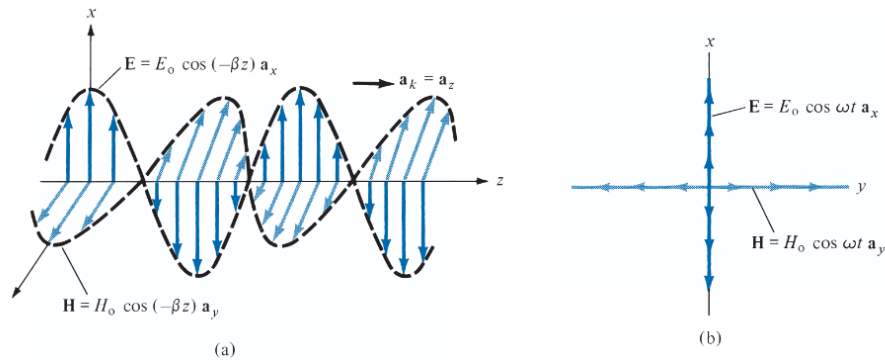


Figure 1.16: Plots of $\mathbf{E}$ and $\mathbf{H}$ (a) as functions of z at $t = 0$; and (b) at $z = 0$. The arrows indicate instantaneous values.

Both $\mathbf{E}$ and $\mathbf{H}$ fields (or EM waves) are everywhere normal to the direction of wave propagation. That means that the fields lie in a plane that is transverse or orthogonal to the direction of propagation. They form an EM wave that has no electric or magnetic field components along the direction of propagation; such a wave is called a *transverse electromagnetic (TEM) wave*. A combination of $\mathbf{E}$ and $\mathbf{H}$ is called a *uniform plane wave* because $\mathbf{E}$ (or $\mathbf{H}$) has the same magnitude throughout any transverse plane, defined by $z = \text{constant}$. The direction in which the electric field points is the polarization of a TEM wave. The wave in eq. (10.29), for example, is polarized in the x -direction. A uniform plane wave cannot exist physically because it stretches to infinity and would represent an infinite energy. Such waves are characteristically simple and fundamentally important. They serve as approximations to practical waves such as those from a radio antenna at distances sufficiently far from radiating sources. Although our discussion after eq. (10.48) deals with free space, it also applies for any other isotropic medium.

PLANE WAVES IN GOOD CONDUCTORS

Plane waves in good conductors comprise another special case of that considered in Section 10.3. A perfect, or good conductor, is one in which $\sigma \gg \omega\epsilon$, so that

$$\frac{\sigma}{\omega\epsilon} \gg 1;$$

that is,

$$\sigma \approx \infty, \quad \epsilon = \epsilon_0, \quad \mu = \mu_0\mu_r \quad (1.155)$$

Hence, eqs. (10.23) and (10.24) become

$$\alpha = \beta = \sqrt{\frac{\omega\mu\sigma}{2}} = \sqrt{\pi f\mu\sigma} \quad (1.156)$$

$$u = \frac{\omega}{\beta} = \sqrt{\frac{2\omega}{\mu\sigma}}, \quad \lambda = \frac{2\pi}{\beta} \quad (1.157)$$

Also, from eq. (10.32),

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma}} = \sqrt{\frac{\omega\mu}{\sigma}} \angle 45^\circ \quad (1.158)$$

and thus $\mathbf{E}$ leads $\mathbf{H}$ by 45° .

$$\mathbf{E} = E_0 e^{-\alpha z} \cos(\omega t - \beta z) \mathbf{a}_x \quad (1.159)$$

then

$$\mathbf{H} = \frac{E_0}{\sqrt{\omega\mu/\sigma}} e^{-\alpha z} \cos(\omega t - \beta z - 45^\circ) \mathbf{a}_y \quad (1.160)$$

Therefore, as the $\mathbf{E}$ (or $\mathbf{H}$) wave travels in a conducting medium, its amplitude is attenuated by the factor $e^{-\alpha z}$. The distance at which the wave amplitude decreases to a factor e^{-1} (about 37% of the original value) is called *skin depth* or *penetration depth* of the medium; that is,

$$E_0 e^{-\alpha\delta} = E_0 e^{-1}$$

or

$$\delta = \frac{1}{\alpha} \quad (1.161)$$

The skin depth is a measure of the depth to which an EM wave can penetrate the medium.

Equation (10.54a) is generally valid for any material medium. For good conductors, eqs. (10.51a) and (10.54a) give

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}} = \frac{1}{\alpha} \quad (1.162)$$

From eqs. (10.51a), (10.52), and (10.54b) for a good conductor,

$$\eta = \frac{1}{\sigma \delta} \sqrt{2} e^{j\pi/4} = \frac{1+j}{\sigma \delta} \quad (1.163)$$

Noting that for good conductors $\alpha = \beta = 1/\delta$, eq. (10.53a) can be written as

$$\mathbf{E} = E_0 e^{-z/\delta} \cos\left(\omega t - \frac{z}{\delta}\right) \mathbf{a}_x$$

showing that δ measures the exponential damping of the wave as it travels through the conductor.

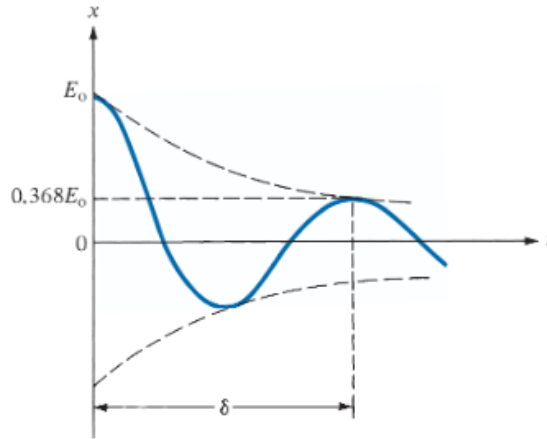


Figure 1.17: Illustration of skin depth.

The phenomenon whereby field intensity in a conductor rapidly decreases is known as the *skin effect*. It is a tendency of charges to migrate from the bulk of the conducting material to the surface, resulting in high resistance. The fields and associated currents are confined to a very thin layer (the skin) of the conductor surface. The skin depth is useful in calculating the ac resistance due to skin effect. The resistance in eq. (5.16) is called the dc resistance, that is,

$$R_{dc} = \frac{\ell}{\sigma S}$$

We define the *surface or skin resistance* R_s (in Ω) as the real part of η for a good conductor. Thus from eq. (10.55)

$$R_s = \frac{1}{\sigma \delta} = \sqrt{\frac{\pi f \mu}{\sigma}} \quad (1.164)$$

Frequency (Hz)	10	60	100	500	10 ⁴	10 ⁸	10 ¹⁰
Skin depth (mm)	20.8	8.6	6.6	2.99	0.66	6.6×10^{-3}	6.6×10^{-4}

*For copper, $\sigma = 5.8 \times 10^7$ S/m, $\mu = \mu_0$, $\delta = 66.1/\sqrt{f}$ (in mm).

Figure 1.18: Skin Depth in Copper

This is the resistance of a unit width and unit length of the conductor. It is equivalent to the dc resistance for a unit length of the conductor having cross-sectional area $1 \times \delta$.

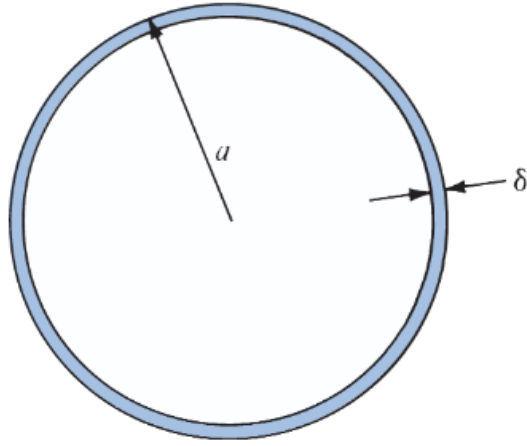


Figure 1.19: Skin depth at high frequencies, $\delta \ll a$

For a given width w and length ℓ , the ac resistance is

$$R_{ac} = \frac{\ell}{\sigma \delta w} = \frac{R_s \ell}{w} \quad (1.165)$$

where $S = \delta w$. For a conductor wire of radius a , $w = 2\pi a$, so

$$\frac{R_{ac}}{R_{dc}} = \frac{\ell/(\sigma 2\pi a \delta)}{\ell/(\sigma \pi a^2)} = \frac{a}{2\delta} = \frac{a}{2} \sqrt{\pi f \mu \sigma}$$

Since $\delta \ll a$ at high frequencies, this shows that R_{ac} is far greater than R_{dc} . In general, the ratio of the ac to dc resistance starts at 1.0 for dc and very low frequencies and increases as frequency increases.

	Lossy Medium	Lossless Medium	Free Space	Conductor
$\alpha =$	$\omega \left[\frac{\mu \epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon} \right)^2} - 1 \right] \right]^{1/2}$	0	0	$\sqrt{\pi f \mu \sigma}$
$\beta =$	$\omega \left[\frac{\mu \epsilon}{2} \left[\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon} \right)^2} + 1 \right] \right]^{1/2}$	$\omega \sqrt{\mu \epsilon}$	$\omega \sqrt{\mu_0 \epsilon_0}$	$\sqrt{\pi f \mu \sigma}$
$\eta =$	$\sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$	$\sqrt{\frac{\mu}{\epsilon}}$	$\sqrt{\frac{\mu_0}{\epsilon_0}} \approx 377$	$(1 + j) \frac{\alpha}{\sigma}$
$\nu = \frac{\omega}{\beta}, \quad \lambda = \frac{2\pi}{\beta}$				

Figure 1.20: Formulas for α , β , η , ν , and λ

POWER AND THE POYNTING VECTOR

As mentioned before, energy can be transported from one point (where a transmitter is located) to another point (with a receiver) by means of EM waves. The rate of such energy transportation can be obtained from Maxwell's equations:

$$\nabla \times \mathbf{E} = -\mu \frac{\partial \mathbf{H}}{\partial t} \quad (1.166)$$

$$\nabla \times \mathbf{H} = \sigma \mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t} \quad (1.167)$$

Dotting both sides of eq. (10.68b) with $\mathbf{E}$ gives

$$\mathbf{E} \cdot (\nabla \times \mathbf{H}) = \sigma E^2 + \mathbf{E} \cdot \epsilon \frac{\partial \mathbf{E}}{\partial t} \quad (1.168)$$

But for any vector fields $\mathbf{A}$ and $\mathbf{B}$,

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$$

Applying this identity to eq. (10.69) (letting $\mathbf{A} = \mathbf{H}$ and $\mathbf{B} = \mathbf{E}$) gives

$$\begin{aligned} \mathbf{H} \cdot (\nabla \times \mathbf{E}) + \nabla \cdot (\mathbf{H} \times \mathbf{E}) &= \sigma E^2 + \mathbf{E} \cdot \epsilon \frac{\partial \mathbf{E}}{\partial t} \\ &= \sigma E^2 + \frac{1}{2} \epsilon \frac{\partial E^2}{\partial t} \end{aligned} \quad (1.169)$$

Dotting both sides of eq. (10.68a) with $\mathbf{H}$,

$$\mathbf{H} \cdot (\nabla \times \mathbf{E}) = \mathbf{H} \cdot \left(-\mu \frac{\partial \mathbf{H}}{\partial t} \right) = -\frac{\mu}{2} \frac{\partial}{\partial t} (\mathbf{H} \cdot \mathbf{H}) \quad (1.170)$$

Hence eq. (10.70) becomes

$$-\frac{\mu}{2} \frac{\partial H^2}{\partial t} - \nabla \cdot (\mathbf{E} \times \mathbf{H}) = \sigma E^2 + \frac{1}{2} \varepsilon \frac{\partial E^2}{\partial t}$$

Rearranging terms and integrating over a volume,

$$\int_v \nabla \cdot (\mathbf{E} \times \mathbf{H}) dv = -\frac{\partial}{\partial t} \int_v \left(\frac{1}{2} \varepsilon E^2 + \frac{1}{2} \mu H^2 \right) dv - \int_v \sigma E^2 dv \quad (1.171)$$

Applying the divergence theorem gives

$$\oint_S (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{S} = -\frac{\partial}{\partial t} \int_v \left(\frac{1}{2} \varepsilon E^2 + \frac{1}{2} \mu H^2 \right) dv - \int_v \sigma E^2 dv \quad (1.172)$$

Equation (10.73) is referred to as *Poynting's theorem*. The quantity $\mathbf{E} \times \mathbf{H}$ is known as the *Poynting vector* $\mathbf{P}$, measured in watts per square meter (W/m²):

$$\mathbf{P} = \mathbf{E} \times \mathbf{H} \quad (1.173)$$

It represents the instantaneous power density vector associated with the EM field at a given point.

Poynting's theorem states that the net power flowing out of a given volume is equal to the time rate of decrease in the energy stored within the volume minus the ohmic losses.

For uniform plane waves,

$$\mathbf{a}_k = \mathbf{a}_E \times \mathbf{a}_H \quad (1.174)$$

Assume

$$\begin{aligned} \mathbf{E}(z, t) &= E_0 e^{-\alpha z} \cos(\omega t - \beta z) \mathbf{a}_x \\ \mathbf{H}(z, t) &= \frac{E_0}{|\eta|} e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta) \mathbf{a}_y \end{aligned}$$

Then

$$\begin{aligned} \mathbf{P}(z, t) &= \frac{E_0^2}{|\eta|} e^{-2\alpha z} \cos(\omega t - \beta z) \cos(\omega t - \beta z - \theta_\eta) \mathbf{a}_z \\ &= \frac{E_0^2}{2|\eta|} e^{-2\alpha z} [\cos \theta_\eta + \cos(2\omega t - 2\beta z - \theta_\eta)] \mathbf{a}_z \end{aligned} \quad (1.175)$$

To determine the time-average Poynting vector:

$$\mathbf{P}_{ave}(z) = \frac{1}{T} \int_0^T \mathbf{P}(z, t) dt \quad (1.176)$$

It can be shown that

$$\mathbf{P}_{ave}(z) = \frac{1}{2} \text{Re}(\mathbf{E}_s \times \mathbf{H}_s^*) \quad (1.177)$$

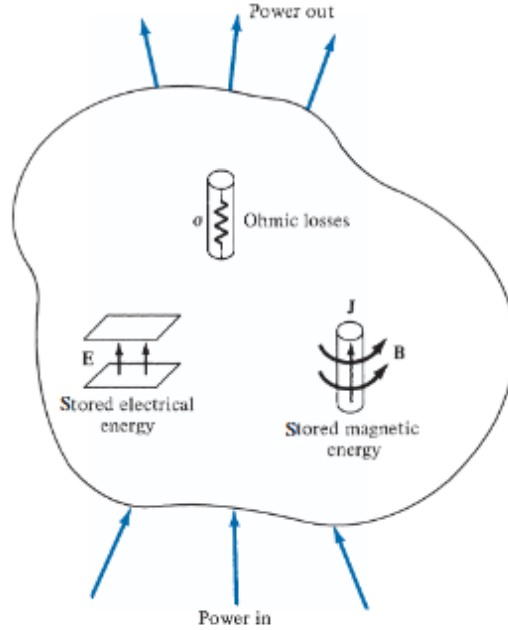


Figure 1.21: Illustration of power balance for EM fields.

Substituting eq. (10.76):

$$\mathbf{P}_{ave}(z) = \frac{E_0^2}{2|\eta|} e^{-2\alpha z} \cos \theta_\eta \mathbf{a}_z \quad (1.178)$$

The total time-average power crossing a surface S is

$$P_{ave} = \int_S \mathbf{P}_{ave} \cdot d\mathbf{S} \quad (1.179)$$

We should note the difference between $\mathbf{P}$, $\mathbf{P}_{ave}$, and P_{ave} : $\mathbf{P}(x, y, z, t)$ is the instantaneous Poynting vector (time varying), $\mathbf{P}_{ave}(x, y, z)$ is its time average (a vector but time invariant), and P_{ave} is the total average power (a scalar).

REFLECTION OF A PLANE WAVE AT NORMAL INCIDENCE

So far, we have considered uniform plane waves traveling in unbounded, homogeneous, isotropic media. When a plane wave from one medium meets a different medium, it is partly reflected and partly transmitted. The proportion of the incident wave that is reflected or transmitted depends on the constitutive parameters $(\varepsilon, \mu, \sigma)$ of the two media involved. Suppose that a plane wave propagating along the $+z$ -direction is incident normally on the boundary $z = 0$ between medium 1 ($z < 0$) characterized by $\sigma_1, \varepsilon_1, \mu_1$ and medium 2 ($z > 0$) characterized by $\sigma_2, \varepsilon_2, \mu_2$.

Incident Wave

$(\mathbf{E}_i, \mathbf{H}_i)$ is traveling along $+a_z$ in medium 1. Suppressing the time factor $e^{j\omega t}$,

$$\mathbf{E}_{is}(z) = E_{i0}e^{-\gamma_1 z}\mathbf{a}_x \quad (1.180)$$

then

$$\mathbf{H}_{is}(z) = H_{i0}e^{-\gamma_1 z}\mathbf{a}_y = \frac{E_{i0}}{\eta_1}e^{-\gamma_1 z}\mathbf{a}_y \quad (1.181)$$

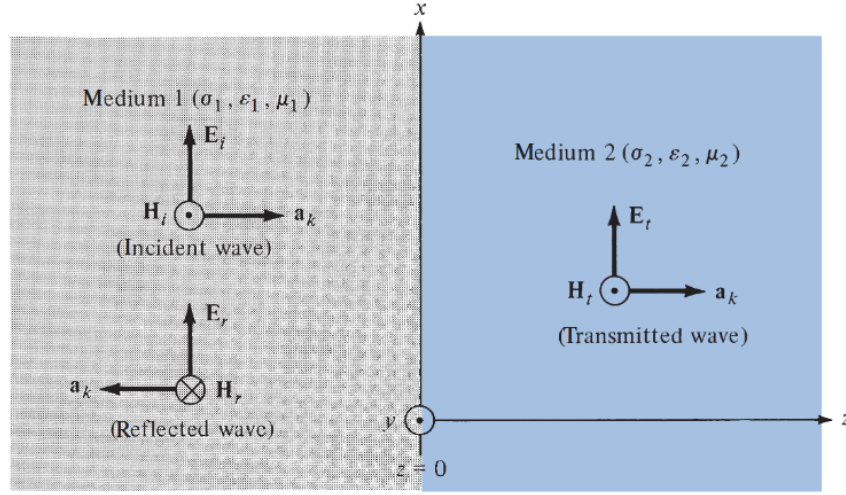


Figure 1.22: A plane wave incident normally on an interface between two different media

Reflected Wave

$(\mathbf{E}_r, \mathbf{H}_r)$ is traveling along $-a_z$ in medium 1.

$$\mathbf{E}_{rs}(z) = E_{r0}e^{\gamma_1 z}\mathbf{a}_x \quad (1.182)$$

then

$$\mathbf{H}_{rs}(z) = H_{r0}e^{\gamma_1 z}(-\mathbf{a}_y) = -\frac{E_{r0}}{\eta_1}e^{\gamma_1 z}\mathbf{a}_y \quad (1.183)$$

Transmitted Wave

$(\mathbf{E}_t, \mathbf{H}_t)$ is traveling along $+a_z$ in medium 2.

$$\mathbf{E}_{ts}(z) = E_{t0}e^{-\gamma_2 z}\mathbf{a}_x \quad (1.184)$$

then

$$\mathbf{H}_{ts}(z) = H_{t0}e^{-\gamma_2 z}\mathbf{a}_y = \frac{E_{t0}}{\eta_2}e^{-\gamma_2 z}\mathbf{a}_y \quad (1.185)$$

At the interface $z = 0$, boundary conditions require continuity of tangential fields:

$$E_i(0) + E_r(0) = E_t(0)$$

$$E_{i0} + E_{r0} = E_{t0} \quad (1.186)$$

$$H_i(0) + H_r(0) = H_t(0)$$

$$\frac{1}{\eta_1}(E_{i0} - E_{r0}) = \frac{E_{t0}}{\eta_2} \quad (1.187)$$

From eqs. (10.87) and (10.88),

$$E_{r0} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} E_{i0} \quad (1.188)$$

$$E_{t0} = \frac{2\eta_2}{\eta_2 + \eta_1} E_{i0} \quad (1.189)$$

We define the reflection coefficient Γ and transmission coefficient τ :

$$\Gamma = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} \quad (1.190)$$

$$E_{r0} = \Gamma E_{i0} \quad (1.191)$$

$$\tau = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2}{\eta_2 + \eta_1} \quad (1.192)$$

$$E_{t0} = \tau E_{i0} \quad (1.193)$$

Note:

1. $1 + \Gamma = \tau$
2. Both Γ and τ are dimensionless and may be complex.
3. $0 \leq |\Gamma| \leq 1$

Standing Wave

For a perfect conductor ($\sigma_2 \rightarrow \infty$),

$$\Gamma = -1, \quad \tau = 0$$

Combining incident and reflected fields:

$$\mathbf{E}_{1s} = (E_{i0}e^{-\gamma_1 z} + E_{r0}e^{\gamma_1 z})\mathbf{a}_x \quad (1.194)$$

Since $E_{r0} = -E_{i0}$,

$$\mathbf{E}_{1s} = -2jE_{i0} \sin \beta_1 z \mathbf{a}_x \quad (1.195)$$

Thus

$$\mathbf{E}_1 = 2E_{i0} \sin \beta_1 z \sin \omega t \mathbf{a}_x \quad (1.196)$$

Similarly,

$$\mathbf{H}_1 = \frac{2E_{i0}}{\eta_1} \cos \beta_1 z \cos \omega t \mathbf{a}_y \quad (1.197)$$

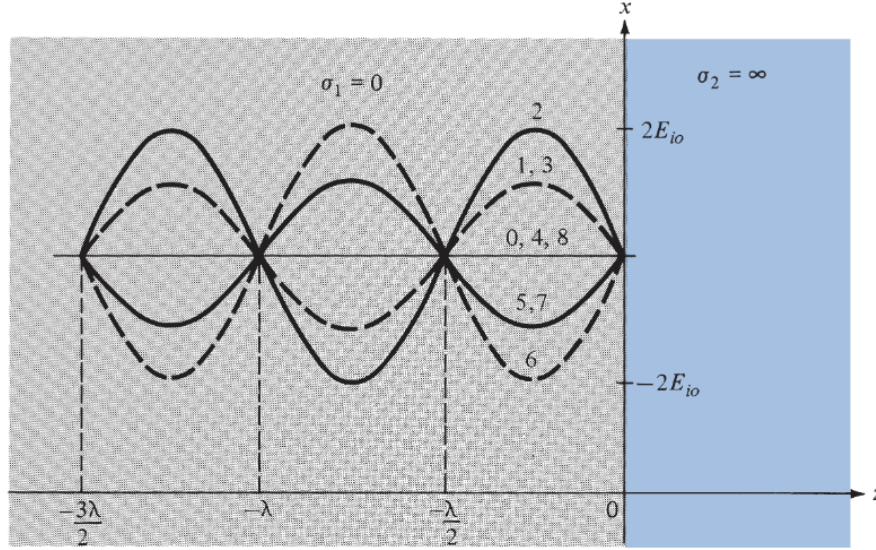


Figure 1.23: Standing waves $E = 2E_{i0} \sin \beta_1 z \sin \omega t \mathbf{a}_x$. The curves 0, 1, 2, 3, 4, ... are, respectively, at times $t = 0, T/8, T/4, 3T/8, T/2, \dots$; $\lambda = 2\pi/\beta_1$.

Case 1: $\eta_2 > \eta_1$

If $\eta_2 > \eta_1$, $\Gamma > 0$.

$$\begin{aligned} -\beta_1 z_{\max} &= n\pi \\ z_{\max} &= -\frac{n\pi}{\beta_1} = -\frac{n\lambda_1}{2}, \quad n = 0, 1, 2, \dots \end{aligned} \quad (1.198)$$

Minimum values occur at

$$\begin{aligned} -\beta_1 z_{\min} &= (2n+1)\frac{\pi}{2} \\ z_{\min} &= -\frac{(2n+1)\pi}{2\beta_1} = -\frac{(2n+1)}{4}\lambda_1, \quad n = 0, 1, 2, \dots \end{aligned} \quad (1.199)$$

Case 2: $\eta_2 < \eta_1$

If $\eta_2 < \eta_1$, $\Gamma < 0$. The positions of maxima and minima are interchanged.

1. $|H_1|$ minimum occurs whenever $|E_1|$ maximum, and vice versa.
2. The transmitted wave in medium 2 is purely traveling (no standing wave).

The ratio of $|E_1|_{\max}$ to $|E_1|_{\min}$ is the standing wave ratio:

$$s = \frac{|E_1|_{\max}}{|E_1|_{\min}} = \frac{|H_1|_{\max}}{|H_1|_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|} \quad (1.200)$$

or

$$|\Gamma| = \frac{s - 1}{s + 1} \quad (1.201)$$

Since $|\Gamma| \leq 1$, we have $1 \leq s \leq \infty$. The standing wave ratio is often expressed in decibels:

$$s_{\text{dB}} = 20 \log_{10} s \quad (1.202)$$